

AI Text Detection Analysis Report

Generated on December 12, 2025 at 10:06 PM

Analyzed File: AI-Powered Tree Cover Detection System Using Satellite Imagery.pdf

Text Source	AI	HUMAN	MIXED
Probability	21.9%	67.3%	10.7%

DETECTION VERDICT

HUMAN-WRITTEN

Confidence: **70.1%**
Uncertainty: 23.5% | Consensus: 87.8%

DETECTION REASONING

- Ensemble analysis indicates with high confidence (70.1%) that this text is **likely human-written** (human probability: 67.4%).
- Metrics show strong consensus among detection methods.
- Uncertainty level: 23.6%.
- Analysis of 1,217 words in **ai_ml** domain using confidence-weighted aggregation with domain calibration ensemble method.

KEY INDICATORS

- Structural** (24.3% weight): High burstiness (0.75) indicates natural variation
- Multi Perturbation Stability** (11.2% weight): Text stability under perturbation (0.00)

CONTENT ANALYSIS

Content Domain	AI ML (14.4% confidence)
Text Statistics	1,217 words 14 sentences
Processing Time	17.16 seconds
Analysis Method	Confidence-Weighted Ensemble Aggregation

METRIC CONTRIBUTIONS

Metric Name	Ensemble Weight (%)
Structural	24.3%
Entropy	6.5%
Perplexity	36.6%
Semantic Analysis	9.3%
Linguistic	12.1%
Multi Perturbation Stability	11.2%

Report continues with detailed metric analysis on the following pages...

DETAILED METRIC ANALYSIS

STRUCTURAL

Analyzes sentence structure, length patterns, and statistical features

Verdict	MIXED	Weight	24.3%
AI Probability	22.4%	Confidence	78.1%

Ai Threshold:	0.570	Avg Sentence Length:	25.07	Avg Word Length:	5.44
Bigram Diversity:	0.905	Burstiness Score:	0.748	Comma Frequency:	0.062
Domain Used:	ai_ml	Human Threshold:	0.430	Length Uniformity:	0.00
Num Sentences:	56	Num Words:	1,358	Punctuation Density:	0.049
Raw Score:	0.400	Readability Score:	21.62	Repetition Score:	0.016
Std Sentence Length:	37.53	Std Word Length:	2.98	Trigram Diversity:	0.981
Type Token Ratio:	0.403	Vocabulary Size:	547		

ENTROPY

Evaluates token diversity and sequence unpredictability

Verdict	MIXED	Weight	6.5%
AI Probability	23.4%	Confidence	42.6%

Ai Pattern Score:	0.500	Ai Threshold:	0.470	Avg Chunk Entropy:	4.14
Char Entropy:	4.19	Domain Used:	ai_ml	Entropy Variance:	0.00
Human Threshold:	0.500	Num Chunks Analyzed:	28.000	Predictability Score:	0.000
Raw Score:	0.460	Sequence Unpredictability:	1.000	Token Diversity:	0.375
Token Entropy:	8.15	Word Entropy:	8.44		

DETAILED METRIC ANALYSIS

PERPLEXITY

Measures text predictability using language model cross-entropy

Verdict	HUMAN	Weight	36.6%
AI Probability	20.1%	Confidence	90.0%

Ai Threshold:	0.510	Avg Chunk Perplexity:	80.09	Avg Sentence Perplexity:	252.75
Cross Entropy Score:	0.80	Domain Used:	ai_ml	Human Threshold:	0.460
Max Sentence Perplexity:	1,140	Min Sentence Perplexity:	41.66	Normalized Perplexity:	0.08
Num Chunks Analyzed:	14.000	Num Sentences Analyzed:	41.000	Overall Perplexity:	53.96
Perplexity Variance:	664.34	Raw Score:	0.220	Std Sentence Perplexity:	241.34

SEMANTIC ANALYSIS

Examines semantic coherence, topic consistency, and logical flow

Verdict	HUMAN	Weight	9.3%
AI Probability	38.6%	Confidence	42.6%

Ai Threshold:	0.570	Avg Chunk Coherence:	0.3383	Coherence Score:	0.3148
Coherence Variance:	0.011	Consistency Score:	0.892	Contextual Consistency:	0.269
Domain Used:	ai_ml	Human Threshold:	0.430	Num Chunks Analyzed:	11.000
Num Sentences:	29	Raw Score:	0.460	Repetition Score:	0.000
Topic Drift Score:	0.710				

DETAILED METRIC ANALYSIS

LINGUISTIC

Assesses grammatical patterns, syntactic complexity, and style markers

Verdict	HUMAN	Weight	12.1%
AI Probability	25.4%	Confidence	54.2%

Ai Pattern Score:	0.000	Ai Threshold:	0.610	Avg Chunk Complexity:	3.572
Avg Sentence Complexity:	3.738	Complexity Variance:	1.552	Domain Used:	ai_ml
Grammatical Consistency:	0.640	Human Threshold:	0.390	Num Chunks Analyzed:	14.000
Num Sentences:	50	Passive Voice Ratio:	0.140	Pos Diversity:	0.009
Pos Entropy:	3.12	Raw Score:	0.350	Syntactic Complexity:	3.099
Transition Word Usage:	0.000	Writing Style Score:	0.463		

MULTI PERTURBATION STABILITY

Tests text stability under perturbation using curvature analysis

Verdict	HUMAN	Weight	11.2%
AI Probability	29.5%	Confidence	87.6%

Ai Threshold:	0.640	Avg Chunk Stability:	0.000	Avg Perturbed Likelihood:	4.145
Curvature Score:	0.145	Domain Used:	ai_ml	Human Threshold:	0.360
Likelihood Ratio:	0.940	Normalized Likelihood Ratio:	0.313	Num Chunks Analyzed:	4.000
Num Perturbations:	10.000	Num Valid Perturbations:	10.000	Original Likelihood:	3.895
Perturbation Variance:	0.048	Raw Score:	0.345	Stability Score:	0.000
Stability Variance:	0.000				

AI Model Attribution

Predicted Model	Gpt-4
Attribution Confidence	10.7%
Domain Used	AI_ML

Model Probability Distribution

Language Model Name	Attribution Probability
Gpt 4	11.5%
Claude 3 Opus	10.4%
Gpt 3.5 Turbo	9.0%
Mixtral	9.0%
Deepseek Chat	5.7%
Gpt 4O	4.9%
Claude 3 Haiku	4.9%
Gpt 4 Turbo	4.9%
Gemini Pro	4.9%
Llama 3	4.9%

Recommendations

- ✓ **Likely human-written:** Consider context and writing history for complete assessment.
- ✓ **Context matters:** Consider author's background, writing history, and situational factors.
- ✓ **Educational approach:** Use detection results as conversation starters about appropriate AI use.
- ✓ **Continuous evaluation:** AI writing evolves rapidly; regular calibration updates maintain accuracy.